# NYC subway disruptions and FHV pickups

*Research methods and findings guide*

Companion to notebook 08 | Revised 45 minute exposure rule

The saved analysis estimates 18.06 additional completed pickups per hour during retained sustained service loss, with a reported 95% interval of 12.06 to 24.05. This guide explains the comparison, the code and the limits of that interpretation.

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# 01 The research question and how to read this guide

I want to understand whether sustained subway service loss is associated with additional completed high-volume FHV pickups in affected NYC taxi zones. The primary estimate in the uploaded saved run is 18.06 additional pickups per hour, with a reported incident-clustered 95% interval from 12.06 to 24.05. It uses 368 episode–zone pairs from 163 incidents.

This guide explains how I reached that result, why each method was used, and where the interpretation can fail. It accompanies the storytelling version of notebook 08. The notebook remains the computational record; this document explains its decisions and code in reading order. The figures and numerical results come from the user's executed notebook, not from a new execution of the local input data during document preparation.

My forecasting work predicts how many pickups might occur. This research instead estimates a contrast: how much pickups change relative to a comparison intended to represent the absence of disruption. Good predictions can help construct a comparison, but predictive accuracy alone cannot establish a causal effect.

The outcome counts completed pickups. It does not count all requests, waiting passengers, unmet demand or identified subway passengers switching to FHVs. A positive estimate is consistent with substitution toward FHVs, but cannot prove individual mode switching.

I read the notebook as a sequence of questions. Can I connect outcomes to exposure without duplicating rows? Which service losses can I measure reliably? What comparison is available? Does the result survive different reasonable choices? Those questions explain the structure of this guide.

The study covers July 1, 2024 through June 30, 2026, with effect estimation beginning September 1, 2024. Each event uses its own preceding eight weeks, ending before its baseline. This is not a single fixed July–August training period. The research is separate from the Citi Bike project and from the MTA construction and production forecasting notebooks.

# 02 Terms used throughout the analysis

A zone-hour is one taxi zone observed during one clock hour. A panel follows many zones across many hours. The joined research panel has 2,890,800 zone-hours across 165 subway-served zones.

An incident is identified by event_id. An episode is identified by episode_key. Related observations can belong to the same incident, so episode or zone counts should not be read as independent incidents. Borough incident counts overlap and cannot be added to obtain the overall incident count.

A spell is an uninterrupted sequence of measured loss bins in one fixed episode, route, direction and station-complex series. A 15-minute bin is a measurement interval, not a passenger outcome interval. Four adjacent bins establish at least 60 continuous minutes of loss.

An episode–zone pair identifies one episode mapped to one taxi zone. This is the unit receiving equal weight in the main average. An incident affecting several zones contributes several pairs. Equal-pair weighting is therefore different from equal-incident weighting or weighting by passenger counts.

Treatment means the measured service-loss exposure used in this research. A control or comparison zone provides simultaneous outcomes intended to represent what might have happened without that exposure. Clean means it passes the implemented recorded-exposure and quality rules; it is not proof that no unrecorded problem occurred.

The baseline is the three complete hours before the earlier warning/exposure anchor. A counterfactual is the outcome that would have occurred without disruption. It cannot be observed for the same zone and hour, which is why a comparison is needed.

A contrast is a calculated difference. An estimand is the quantity the design aims to describe: here, the equal-pair average before-to-during change in the adjusted pickup gap among retained pairs. A coefficient is a fitted regression parameter; the coefficient on during is 18.056784 before rounding.

A fixed effect gives a group its own baseline. Clustering adjusts uncertainty for dependence within specified groups. Sensitivity analysis asks how results change under alternative choices. None of these terms means the causal assumptions have been proven.

# 03 Inputs and protecting the outcome panel

The verified disruption source is data/silver/mta_disruptions/full_period_3_5_20261012_034129 under the project root. Its manifest records construction settings. The outcome and feature source is data/gold/zone_hour_pickups_weather_sports_concerts.parquet. Zones 1, 104 and 105 are excluded.

load_research_panel() reads the inputs, checks the intended period and joins MTA exposure with pickups by LocationID and hour_local. The saved join retains all 2,890,800 MTA zone-hours, with unique keys and zero unmatched outcomes. This checks the structural join, not the correctness of every exposure record.

The code uses UTC for interval arithmetic and converts to New York local time for clock labels and research scope. Daylight-saving time creates repeated or nonexistent local labels, so a seemingly ordinary local-hour key can be problematic. The engineering pipeline replaced some repeated November-hour pickup counts with neighboring-week values. Those are not observed research outcomes and are excluded, along with nonexistent spring-forward hours.

apply_research_quality() keeps underlying rows and creates flags for research eligibility. Keeping flags makes the reason for exclusion visible and avoids treating an absent or unknown exposure as zero. A row can fail more than one rule, so exclusion counts cannot always be summed.

For censored episodes, the saved three-hour endpoint reflects a processing rule. It does not demonstrate that service returned. Their mapped zone-hours are blocked from the first recorded incident alert through 48 hours after the last active update. This finite buffer is a working research assumption.

The buffer covers 522,876 zone-hours, or 18.09% of the panel. Manhattan's share is 24.78%, compared with 17.28% in Brooklyn, 15.16% in Queens and 14.27% in the Bronx. These differences can change which locations and histories remain available. The buffer is not measured recovery and does not justify an after-recovery interpretation.

# 04 Constructing loss spells and the revised exposure rule

screen_episodes() restricts saved episodes by onset dates, nonholiday weekday daytime onset, selected causes, observed closing update, successful measurement, nontruncated windows and no recorded nearby alert gap. The saved screening leaves 2,605 candidate episodes. Passing this screen is only the beginning of treatment construction.

construct_treatment_hours() selects measured lost or partial_lost bins, sorts by episode, route, direction, station complex and bin_utc, and starts a new spell whenever consecutive timestamps differ from 15 minutes. It checks that elapsed duration agrees with bin count. Access loss is checked as contained within route loss. Diversion alone does not qualify.

The original approach retained complete clock hours within qualifying spells. We revised it because 45 minutes of substantial loss could be discarded at the edge of a longer spell. The implemented rule still requires a spell lasting at least 60 continuous minutes and containing at least one complete clock hour. From that same spell, a clock hour can enter when its overlap is at least 45 minutes.

For a 10:15–12:15 spell, 10:00–11:00 contributes 45 minutes and 11:00–12:00 contributes 60. Both can enter. The 12:00–13:00 hour contributes 15 and is excluded. A 10:15–11:15 spell remains excluded because no full clock hour lies inside it. A standalone 45-minute spell is excluded too.

For each candidate clock hour, the code finds the later of hour start and spell start, and the earlier of hour end and spell end. Their difference gives overlap minutes. It never adds separate short spells or station series together to reach 45 minutes. Duplicate episode–zone–hour keys are then removed.

Measurement-window boundary and shared-attribution flags remain exclusions. A boundary hour in the clock-hour sense is different from a spell touching the edge of its measurement window. The revised rule changes only the former. Pickups remain whole-hour counts; no 45/60 scaling or per-minute effect is estimated.

# 05 Baseline construction and the sample funnel

prepare_event_candidates() maps all attributed episode flags to zone-hours, including noncandidate episodes. This prevents excluded causes or other recorded incidents from silently becoming clean treatment context. Candidate treatment hours are removed when other measured episodes or recorded background route loss overlap.

The baseline anchor is the earlier of first attributed zone exposure and incident_start_uncapped, the first recorded incident alert. Using the earlier time avoids treating a warning or early exposure as an unaffected baseline. The code floors that anchor to the hour and takes the preceding three complete hours.

All three baseline hours must have usable outcomes and quality, no recorded announced exposure, no background loss and no measured episode in that zone-hour. Missing panel coverage is unusable. Baseline hours may fall outside daytime, because a morning disruption still needs preceding clean hours.

The revised 45-minute rule initially raises screened treatment hours from 2,201 to 2,430, adding 229 or 10.4%. After overlap and baseline checks there are 679 candidate pairs, 226 incidents and 947 treatment hours. The earlier full-hour run had 666 candidate pairs.

Comparison geography and event-window screening leave 676 of 679 pairs with at least one clean candidate comparison. Historical matching is more restrictive and retains 370 pairs from 164 incidents, with 1,110 baseline observations and 514 treatment observations. The baseline count equals 370 times three.

Historical weekday–hour support subsequently excludes two Brooklyn pairs. The primary sample is therefore 368 pairs from 163 incidents. These stages have different denominators and should not be mixed.

The net gain from 461 to 514 matched treatment hours is 53 hours or 11.5%. This is not necessarily a count of 53 identical-pair additions: expanded windows can change eligibility and selected matches. A strict decomposition would require tracking membership, which was not needed for the focused revision.

# 06 Choosing simultaneous comparison zones

screen_comparison_zones() generates same-borough comparisons and removes the treated zone itself. It also removes zones sharing a station complex and any zone with attributed flags from the same episode. A zone's own earlier history remains available; only self as a simultaneous untreated comparison is prohibited.

For each pair, the comparison window runs from the first baseline hour to the last retained treatment hour, including intervening hours. Every hour must have known usable quality, no recorded announced exposure, no recorded episode and no background route loss. The expanded treatment window therefore requires a fresh comparison screen.

score_comparison_candidates() uses the earlier eight-week window, ending before baseline. Treated and comparison pickups must be observed at identical usable historical timestamps. At least 280 common hours and 50% coverage are required. The mean comparison/treated volume ratio must lie between 0.5 and 2.0.

Root mean squared error, or RMSE, is the square root of the average squared difference in historical counts. It penalizes large count differences more heavily than small ones. Here it ranks historical resemblance in the pickup-count units used by the effect estimate. It is not a score of causal validity.

select_matches() retains up to three eligible comparisons. Their outcomes are averaged equally within each event hour. Matching uses historical outcomes, not disruption-hour pickups. Keeping selection separate from scoring lets the adjacency check reuse the same historical scores.

Overall retention is 370 of 679 pairs, or 54.49%; 252 pairs have three matches. Brooklyn retention is 66.06%, Queens 63.87%, Bronx 45.95% and Manhattan 37.26%. The sample is selective, particularly in Manhattan.

Matched and unmatched groups have median baseline pickups of 117.2 and 121.0 overall and similar mean treatment duration of about 1.4 hours. These summaries do not eliminate selection concerns. Matching balances selected observed patterns, not all unobserved event-time determinants.

# 07 Calculating the difference in differences

build_did_sample() combines baseline and retained treatment hours, attaches treated and comparison outcomes, and checks stable comparison membership. The pickup gap is treated_total_pickups minus comparison_total_pickups. Comparison totals here are equal-weight means across selected zones, not sums.

calculate_pair_contrasts() first averages each outcome within each pair and period. It subtracts before from during for treated pickups and comparison pickups, then subtracts those two changes. An assertion verifies that this is the same as during gap minus baseline gap.

An illustrative treated increase from 100 to 150 and comparison increase from 80 to 100 gives (150 minus 100) minus (100 minus 80), or 30 pickups per hour. This removes a shared 20-pickup increase, but assumes the comparison change credibly represents the treated zone's absent-disruption change.

In the saved run, the average treated change is 45.35 and the average comparison change is 28.02. Their difference is 17.33 pickups per hour. The raw interval is 10.84 to 23.82. Looking only at the treated change would ignore a substantial simultaneous increase in comparison zones.

The parallel-change assumption concerns how treated and comparison outcomes would have changed without treatment. It is not a requirement that their pickup levels be identical. A stable level gap can disappear when changes are taken; a changing untreated gap can bias the contrast.

Each pair is weighted equally after averaging its hours. A three-hour treatment does not get three times the weight of a one-hour treatment. Multiple pairs from one incident still contribute multiple times, which is why the estimand is pair-weighted.

The raw result remains a sensitivity reference. The primary analysis additionally removes usual weekday–hour gaps and fits weather terms. These choices address specified concerns but do not eliminate all confounding, spillovers or exposure errors.

# 08 Removing normal weekday and hour differences

A treated zone and its comparison can have different daily demand curves even when their average volumes are similar. Baseline and treatment hours often lie at different clock times, so the raw gap can change without a disruption.

check_weekday_hour_support() identifies all weekday–hour combinations needed by the event observations, including early baseline hours. It counts earlier clean dates shared by the treated zone and every selected comparison. Historical completeness here is stricter than matching one comparison at a time: all selected zones must contribute simultaneously.

adjust_historical_gaps() keeps selected comparisons fixed. It estimates the mean treated-minus-comparison gap for each weekday–hour in the earlier eight-week window and subtracts that profile from the observed hourly gap. This fitted historical profile never uses the event's pickup outcomes.

For an example, suppose the actual gap is 20 before and 50 during, while the usual gap for those times is 15 and 25. The adjusted gaps are 5 and 25, giving a 20-pickup contrast instead of the raw 30. These numbers illustrate the operation, not observed pair results.

The support check retains 368 of 370 pairs. Two Brooklyn pairs lack at least one required combination and are excluded instead of filled. The typical pair's weakest combination has only three clean dates, and some have one. Nonmissing history is not necessarily precise history.

The weekday–hour-only contrast is 18.40 pickups per hour, compared with 17.33 raw. Because the samples differ by two pairs, their difference is not purely the effect of subtracting profiles. The historical adjustment also adds estimation uncertainty that the reported downstream intervals do not fully propagate.

This method does not estimate a separate causal effect for each clock hour. It removes an estimated normal gap pattern so the remaining before-to-during comparison is less sensitive to ordinary daily differences.

# 09 Weather regression and the primary coefficient

The weather model starts from the historically adjusted gap. It fits that outcome as a pair-specific baseline plus a common during coefficient plus weather contributions and residual variation. The during indicator equals zero in baseline hours and one in retained treatment hours.

There are eight weather predictors: rain, snowfall, wind and temperature for the treated zone and the corresponding averages across selected comparison zones. Separate coefficients allow treated and comparison weather to have different relationships with the gap. Those coefficients are shared across pairs. Matched DiD does not include the sports or concert flags.

add_weather_terms() scales rain by 0.1 inches, snow by 1 inch, wind by 5 mph and temperature by 10 degrees Fahrenheit. In this unpenalized linear model, scaling changes coefficient units rather than fitted predictions.

fit_weather_did() uses weighted least squares. The weight for each observation is one divided by the number of observations in its pair and period. Each pair thus contributes total weight one before and one during. The fitted parameters minimize the weighted sum of squared residuals.

Suppose a pair's historically adjusted contrast is 20 and the fitted weather contribution increases by 3 between periods. Its remaining contrast is 17. If the weather contribution decreases by 3, the remaining contrast is 23. Weather adjustment can move results in either direction. These numbers are illustrative.

pair_weather_adjusted_contrasts() subtracts fitted weather contributions, averages within each period and differences the periods. The code asserts that the mean of these pair contrasts equals the fitted common during coefficient, 18.056784 in the saved run.

The main reported estimate is 18.06 with incident-clustered interval 12.06 to 24.05. Weather coefficients and the during coefficient are estimated jointly from the short event windows. This is different from learning weather responses only on earlier clean history. We discussed that alternative but did not implement it.

# 10 Weather limitations and why the current method remains

The weather terms assume linear additive responses. The model has no snow threshold, temperature curve, weather-by-disruption interaction or pair-specific weather slope. Different neighborhoods can respond differently to the same storm, beyond what shared coefficients describe.

The early December snow example explains historical support. If an event's preceding eight weeks contain no snowfall, those observations cannot identify a historical snow response. A small amount of snow also provides weak support for predicting an intense storm. Many zones observed during one storm do not create many independent storms.

The primary weather regression uses baseline and disruption windows across events, rather than fitting only an event's eight-week history. It may obtain snow variation elsewhere in the retained event observations. Nevertheless, weather that closely coincides with disruption can be hard to separate from the during indicator. Full numerical rank does not establish reliable weather support.

Simultaneous comparisons help because treated and comparison zones experience the same actual times. If both respond similarly to snow, some common movement cancels in the gap. This is helpful but not automatic: residential and business zones may have different storm responses, and local supply conditions can differ.

Weather-only adjustment moves 17.33 to 17.46. After historical adjustment, weather moves 18.40 to 18.06. These modest changes show stability to this fitted specification. They do not prove that weather has no effect or is completely controlled.

We considered learning weather from earlier clean history, pooling responses, and using more flexible models. We retained the current matched linear adjustment to avoid adding complexity without a demonstrated need. No new historical pooled weather estimator was fitted.

A causal reading still requires credible comparison changes, accurate enough exposure, and limited remaining confounding and spillovers. Calendar events, localized travel shocks, weather measurement error and driver relocation can remain. For this reason the report says the evidence is consistent with increased pickups under the study assumptions.

# 11 Uncertainty and borough interpretation

Several pairs can belong to one incident, so treating every pair as independent would give misleading uncertainty. The primary fitted regression clusters by event_id, allowing dependence within incidents. Finite-sample correction and t-based inference are requested in the regression code.

Two-way clustering also groups observations by treated LocationID. It retains the same 18.06 coefficient and gives interval 12.33 to 23.78. A second clustering dimension does not necessarily widen an interval. These choices do not fully address shared comparison zones, related incidents or all calendar dependence.

The primary regression interval includes uncertainty from estimating parameters within that regression, conditional on the supplied historical adjustment and sample. It does not fully propagate historical-profile estimation, matching selection, disruption measurement or model-choice uncertainty.

Borough-varying regression estimates are Bronx 12.42 [2.81, 22.03], Brooklyn 22.11 [11.98, 32.24], Manhattan 10.21 [minus 1.19, 21.62], and Queens 19.73 [7.21, 32.24]. Its pair-share-weighted overall contrast is 18.17, a different specification from the primary common coefficient.

The other borough table averages recovered primary pair contrasts and treats fitted weather coefficients as fixed. Its intervals can be narrower. Manhattan's lower bound of 0.12 in that summary does not override the borough regression interval including zero. The main overall interval remains 12.06 to 24.05, not the narrower fixed-adjustment pair-summary interval.

A 95% confidence interval is not a range covering 95% of individual effects and is not proof that the assumptions hold. An interval including zero is inconclusive; it does not establish no effect. Comparing overlapping intervals is not a formal test of differences between boroughs or models.

Brooklyn and Queens are the most consistent borough findings across the matched and same-zone analyses. Bronx depends on comparison method. Manhattan remains inconclusive. No formal borough-difference test was implemented.

# 12 The event study and what the curve means

build_event_study_panel() constructs relative hours minus 6 through plus 3 around the earlier warning/exposure anchor, floored to the hour. The code checks that minus 3 through minus 1 match the baseline hours. Each pair's adjusted gap at minus 1 is subtracted from its other hours.

The balanced sample has usable observations at every relative hour, keeping membership constant. It contains 201 of 368 pairs, or 54.6%. The unbalanced version includes all usable pairs at each hour, ranging from 271 to 368 in this run. It uses more data but changes composition across hours.

Pre-anchor treated hours must pass the baseline-style clean checks. Comparison zones must remain clean at every used hour. Post-anchor treated hours need usable quality, but they are not required to satisfy the primary 45-minute treatment definition. Historical profiles and weather coefficients are reused from the primary analysis.

All reported pre-anchor intervals include zero. Balanced pre-anchor estimates range from minus 0.33 to 2.92. This shows no clear prior divergence at the reported precision, but no joint pretrend test is implemented and absence of significance does not prove parallel trends.

At plus 1, the balanced estimate is 20.26 [11.44, 29.09], and the unbalanced estimate is 17.83 [11.64, 24.03]. Both decline afterward. At plus 3, they are 5.16 and 4.34 respectively.

This is a timing diagnostic around the anchor, not a dynamic effect measured from confirmed loss onset or recovery. Hour zero can precede measured loss, and later hours can include active loss, transition or other conditions. The decline cannot be called a recovery response.

We discussed adding a loss-aligned curve but did not implement it. Simply moving the reference to the first full treatment hour and using its preceding hour as a clean baseline would be unsafe because that preceding hour might already contain warning or partial loss. The current curve is retained with a precise interpretation.

# 13 Placebo dates and nearby comparisons

build_multidate_placebo() shifts each full comparison window by whole weeks. This preserves weekday and clock hour. Backward shifts are 9–12 weeks; forward shifts are 1–12 weeks. The matching history is excluded to avoid evaluating the same period used to select comparisons.

Every shifted treated and comparison hour must pass the clean and holiday checks. The selected zones, historical gap profile and primary weather coefficients remain fixed. Using future placebo dates is acceptable for this retrospective diagnostic, but it does not represent a prospective prediction exercise.

Placebo coverage is 338 of 368 pairs, or 91.8%, with a median of three dates. On those same pairs, real events give 18.02 [12.76, 23.28], placebo means 1.62 [minus 0.53, 3.76], and real minus placebo 16.41 [10.80, 22.01]. These show substantial separation while keeping the comparison sample aligned.

placebo_randomization() draws one date per pair and averages the contrasts, repeated 2,000 times. The saved value is 0.0005, not zero; the table rounds to 0.000. Independent pair draws ignore shared incidents, zones and dates, so this is a descriptive distribution rather than a formal randomization test. Its central percentiles are not a confidence interval for the primary effect.

For the spillover check, load_zone_adjacency() reads data/external/tlc_zones.geojson, renames lowercase locationid and converts it to integer. Projection EPSG 2263 measures distances in feet. Both polygons are buffered by 50 feet, so gaps up to roughly 100 feet can count, rather than exactly 50 feet.

Of 368 primary pairs, 101 use at least one adjacent comparison. On the identical supported set of 361 pairs from 159 incidents, original matches give 18.21 [12.14, 24.28] and nonadjacent matches give 18.08 [11.88, 24.28]. This small difference is reassuring for this restriction but does not rule out distant spillovers or identify driver and passenger mechanisms.

# 14 Influence checks and the relative percentage

The leave-one-incident-out calculation removes each incident's pair contrasts and recomputes the mean. With 163 incidents, there are 163 such means. Fitted profiles, weather coefficients and matches remain fixed; this is not an end-to-end refit after each removal.

The resulting range is 16.94 to 18.77. Incident 201488 is most influential under the largest absolute change criterion. No individual incident eliminates the positive average in this fixed-adjustment check.

The mean pair contrast is 18.06, median 11.93, and 10% trimmed mean 13.94. Trimming removes the lowest 10% and highest 10% of contrasts. Larger positive values pull up the mean, but the central descriptive summaries remain positive. No interval for the trimmed mean is calculated here.

The percentage constructs expected_during as observed during-period mean pickups minus primary_contrast. It sums pair contrasts and divides by the sum of those expected values, multiplied by 100. This is a ratio of pair-level sums, not an average of individual pair percentages and not an hour-weighted citywide pickup increase.

The estimate is 11.23%, with a reported incident-bootstrap interval of 8.20% to 14.40%. The bootstrap resamples incident-level sums, preserving the incident's included pairs within a draw. It keeps fitted adjustments fixed and does not reproduce matching or weather estimation in each draw.

Two expected pickup values are negative, an impossible count. Subtracting a contrast is not a constrained count-prediction model, so such values can occur. These two values are also the pairs counted below five per hour. The aggregate ratio includes them as implemented; it should be described as approximate.

I retain pickups per hour as the primary reporting unit. The percentage helps convey practical size, but is secondary and should not be presented as a precise universal percentage increase for every zone or disruption.

# 15 Same zone linear and XGBoost references

prepare_same_zone_data() collects clean historical observations and event rows for each primary pair. Histories end before baseline. The linear model includes weekday–hour baselines plus rain, snowfall, wind, temperature, unofficial-holiday and recorded sports/concert-window flags. Official holidays remain excluded.

fit_linear_reference() subtracts weekday–hour means from predictors and pickups, estimates within-group slopes using np.linalg.lstsq, and restores the target weekday–hour baseline for prediction. This is equivalent to an OLS fixed-effect formulation for the supported predictions. The projection check flags target predictor directions the historical design cannot estimate.

All 368 pairs pass the implemented event prediction-support check. That does not establish accuracy, abundant snow examples or absence of extrapolation. Event flags use estimated windows, not direct crowd counts. No pickup lags, fares or waiting times enter these models.

fit_same_zone_reference() computes observed minus predicted pickups, averages by pair and period, and differences during minus before. Its overall contrast is 14.81 [7.94, 21.68]. Bronx is 6.52 [minus 4.72, 17.77], Brooklyn 21.22 [11.14, 31.30], Manhattan 1.37 [minus 10.65, 13.40], and Queens 17.37 [3.74, 31.00]. Intervals omit full prediction-model fitting uncertainty.

The XGBoost appendix splits the earlier eight weeks into six for training and two for validation. Four settings combine max_depth 3 or 5 with min_child_weight 5 or 20. Trees remain 200, learning rate 0.05, and regularization 10. This current appendix is not the earlier 18-configuration experiment.

Each pair selects its own validation setting, then refits on its full history. Selected validation scores are optimistic and histories overlap. Linear validation MAE/RMSE are 20.60/35.14, versus XGBoost 21.34/35.75. These summarize pair errors, not an independent test set.

XGBoost's overall contrast is 14.53 [7.80, 21.27]. Brooklyn and Queens remain positive while Bronx and Manhattan intervals include zero. I keep linear as the simpler historical reference and stop tuning. The GPU fallback warning concerns prediction speed and input-device mismatch; saved training reports cuda:0.

# 16 Decisions findings and remaining work

The primary result is 18.06 pickups per hour [12.06, 24.05]. Raw matched DiD is 17.33, weather-only 17.46, weekday–hour-only 18.40, borough-varying overall 18.17, same-zone linear 14.81 and XGBoost 14.53. The adjacency-restricted comparison is 18.08 on its smaller common sample.

These findings consistently point toward a positive overall contrast, but share observations and potential biases. Their range is not a confidence interval, and their average would not create a better causal estimate. Different samples and specifications also prevent interpreting every numerical difference as the isolated contribution of one adjustment.

The earlier full-hour primary estimate was 18.08. The revised rule retains more exposure but gives 18.06, so the substantive overall conclusion is stable to this change. We did not estimate a causal dose response for 45 versus 60 minutes. The treatment remains a qualifying hourly category.

The original checkpoint table deliberately preserves old values. Panel rows agree; candidates, matches, hours and estimates differ as expected. False in matches is numerical disagreement, not automatically a validation failure. The old weekday–hour row label mentions 366 pairs, while the new current value uses 368; the revised notebook explains that stale label without changing code or saved output.

Recovery analysis remains planned. We must first determine whether saved tables contain usable service evidence after loss ends. Recovery cannot be inferred solely from an alert end or a missing attributed record. A future design must handle intermittent loss, different station and route recovery times, new nearby disruptions and comparison eligibility through the extended horizon.

The likely future target is lingering pickup changes after observed recovery relative to no disruption. That differs from estimating the effect of restoring service compared with continuing disruption. No recovery horizon, sample or result has been estimated here. Dropoffs, unmet demand and individual switching also remain outside scope.

For the present study, I retain adjusted matched DiD as primary, report Brooklyn and Queens as most consistent across methods, treat Bronx as method-sensitive and Manhattan as inconclusive, and explain the remaining assumptions directly.

# 17 Code reading and reproducibility

Read configuration first, then follow function definitions and their calls. The source notebook has saved outputs from the user's clean rerun. The storytelling copy changes Markdown only; all code cells, outputs and execution counts are preserved. Interpretation numbers are tied to this saved run and must be reviewed after a future rerun.

load_research_panel and apply_research_quality establish keys and eligible panel hours. screen_episodes and construct_treatment_hours define the measured exposure. prepare_event_candidates creates overlap checks, baselines and historical boundaries. screen_comparison_zones, score_comparison_candidates and select_matches identify comparisons.

build_did_sample and calculate_pair_contrasts build hourly gaps and raw pair changes. check_weekday_hour_support and adjust_historical_gaps estimate the normal gap. add_weather_terms, fit_weather_did and pair_weather_adjusted_contrasts produce the primary estimate and diagnostic pair contrasts.

summarize_borough_model extracts regression coefficients and the weighted overall contrast. build_event_study_panel and estimate_event_study calculate relative-hour curves. build_multidate_placebo and placebo_randomization construct clean-date diagnostics. load_zone_adjacency and estimate_primary rerun the specification with restricted comparisons.

prepare_same_zone_data, fit_linear_reference and fit_same_zone_reference create the historical linear sensitivity analysis. run_xgboost_appendix performs the four-configuration chronological validation experiment. mean_row and model_row organize summaries; display_table controls presentation rather than estimation.

merge(validate=...) checks expected join relationships. drop_duplicates prevents repeated exposure records from multiplying outcomes. groupby aggregates at a stated unit; its choice determines weighting. Assertions check keys, chronology, membership, completeness, continuity and model rank. They guard implementation, not every research assumption.

To reproduce, use the ride-demand kernel, confirm project paths and GeoJSON schema handling, restart and run from the top. Do not change configuration midway. If skipping XGBoost, clear its stale outputs. The event-study NumPy timedelta deprecation warning needs a later maintenance fix; it did not stop this saved run. The CPU/GPU prediction warning similarly did not stop XGBoost.

This write-up introduces no new fitted model, source data edits or external results. It documents the completed saved analysis and the design decisions discussed during its revision.